A six-point counterexample to the idempotent-measure projectivity conjecture

Solution packet for the conjecture in arXiv:2203.09765

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Full negative answer. The conjecture fails already for the finite group S_3. An idempotent measure μ of norm 2 generates exactly the same left ideal of $\ell^1(S_3)$ as a norm-one idempotent q. The ideal is therefore metrically projective, although the unique density of μ has norm 2, not 1.

The conjecture

After Proposition 3.2.4, Nemesh conjectures that for an idempotent measure $\mu \in M(G)$ the left ideal $I = L^1(G) * \mu$ is metrically projective if and only if $\mu = p\, m_G$ for some $p \in I$ with $\|p\|_1 = 1$ [1].

> are of the form $L_1(G) * \mu$ for some idempotent measure μ. In fact, this class of ideals in case of commutative compact groups G coincides with those left ideals of $L_1(G)$ that admit a right bounded approximate identity.
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> **Proposition 3.2.4.** *Let G be a locally compact group and $\mu \in M(G)$ be an idempotent measure, that is $\mu * \mu = \mu$. If the left ideal $I = L_1(G) * \mu$ of Banach algebra $L_1(G)$ is topologically projective $L_1(G)$-module, then $\mu = pm_G$, for some $p \in I$.*
>
> *Proof.* Let $\phi : I \to L_1(G)$ be arbitrary morphism of left $L_1(G)$-modules. Consider $L_1(G)$-morphism $\phi' : L_1(G) \to L_1(G) : x \mapsto \phi(x * \mu)$. By Wendel's theorem [[66], theorem 1], there exists a measure $\nu \in M(G)$ such that $\phi'(x) = x * \nu$ for all $x \in L_1(G)$. In particular, $\phi(x) = \phi(x * \mu) = \phi'(x) = x * \nu$ for all $x \in I$. It is now clear that $\psi : I \to I : x \mapsto \nu * x$ is a morphism of right I-modules satisfying $\phi(x)y = x\psi(y)$ for all $x, y \in I$. By paragraph (ii) of lemma 2.1.15 the ideal I has a right identity, say $e \in I$. Then $x * \mu = x * \mu * e$ for all $x \in L_1(G)$. Two measures are equal if their convolutions with all functions of $L_1(G)$ coincide [[25], corollary 3.3.24], so $\mu = \mu * e m_G$. Since $e \in I \subset L_1(G)$, then $\mu = \mu * e m_G \in M_a(G)$. Set $p = \mu * e \in I$, then $\mu = pm_G$. $\square$
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> We conjecture that the left ideal $L_1(G) * \mu$ for idempotent measure μ is metrically projective $L_1(G)$-module iff $\mu = pm_G$ where $p \in I$ and $\|p\| = 1$.

Idea of the counterexample

Metric projectivity belongs to the module I, whereas the proposed norm condition belongs to the particular idempotent measure used to present I. These data are not invariant together. If q is an idempotent and $x \in (1-q)Aq$, then

$$qx = 0, \qquad xq = x, \qquad x^2 = 0.$$

Consequently $\mu = q + x$ is another idempotent, and $A\mu = Aq$. In a noncommutative group algebra one can choose q with norm one while arranging that q and x have disjoint supports, forcing $\|\mu\|_1 > 1$.

The explicit example

Let $G = S_3$, with products composed from right to left, and identify $M(G) = L^1(G) = \ell^1(G)$ using counting Haar measure. Put

$$s = (12), \qquad a = \delta_{(23)}, \qquad q = \frac{1}{2}(\delta_e + \delta_s), \qquad x = (\delta_e - q) * a * q,$$

and set $\mu = q + x$. Direct expansion gives

$$x = \frac{1}{4}\big(\delta_{(23)} + \delta_{(132)} - \delta_{(123)} - \delta_{(13)}\big). \tag{1}$$

Theorem 1. *The measure μ is idempotent, the left ideal $I = \ell^1(S_3) * \mu$ is a metrically projective $\ell^1(S_3)$-module, and μ has no L^1-density of norm one. Hence the conjectured "only if" direction is false.*

Proof. Since q is the normalized Haar measure of the subgroup $\{e, (12)\}$, it is an idempotent and $\|q\|_1 = 1$. The definition of x immediately yields

$$q * x = 0, \qquad x * q = x,$$

and

$$x * x = (\delta_e - q) * a * q * (\delta_e - q) * a * q = 0$$

because $q * (\delta_e - q) = 0$. Therefore

$$\mu * \mu = q * q + q * x + x * q + x * x = q + x = \mu.$$

The same identities give

$$\mu * q = \mu, \qquad q * \mu = q. \tag{2}$$

The first relation in (2) implies $\ell^1(G)\mu \subseteq \ell^1(G)q$, and the second gives the reverse inclusion. Thus

$$I = \ell^1(G)\mu = \ell^1(G)q.$$

Since q is a norm-one idempotent, Proposition 2.1.13(i) of the source says that $\ell^1(G)q$ is metrically projective. Equivalently, it is the contractive retract of the free rank-one left module $\ell^1(G)$ given by right convolution with q and the inclusion of its range.

The supports of q and x in (1) are disjoint, so

$$\|\mu\|_1 = \|q\|_1 + \|x\|_1 = 1 + 1 = 2.$$

For any choice of Haar normalization, the L^1 norm of the Radon–Nikodym density of μ is its total-variation norm, namely 2. That density is unique up to Haar-null sets. Hence there is no p with $\mu = p\, m_G$ and $\|p\|_1 = 1$, even though I is metrically projective. $\square$

A general family

The phenomenon is not special to the six-element computation.

Proposition 1. *Let G be a discrete group with a finite nonnormal subgroup H. Write q for normalized Haar measure on H, and choose $g \notin N_G(H)$. Then*

$$x = (\delta_e - q) * \delta_g * q, \qquad \mu = q + x$$

give an idempotent $\mu \in \ell^1(G)$ such that

$$\ell^1(G)\mu = \ell^1(G)q, \qquad \|q\|_1 = 1, \qquad \|\mu\|_1 > 1.$$

Thus every such group supplies a counterexample to the conjecture.

Proof. The Peirce relations used in Theorem 1 are formal and give idempotency and equality of the two left ideals. If $x = 0$, then $q*\delta_g*q = \delta_g*q$, so the right coset gH is invariant under left multiplication by H. Hence $g^{-1}Hg \subseteq H$, and finiteness gives $g^{-1}Hg = H$, contrary to $g \notin N_G(H)$. Thus $x \neq 0$. Its support is contained in the double coset HgH, which is disjoint from H; therefore $\|\mu\|_1 = \|q\|_1 + \|x\|_1 > 1$. □

Adversarial checks

1. The inclusions of principal *left* ideals have the correct orientation: $\mu q = \mu$ gives $A\mu \subseteq Aq$, while $q\mu = q$ gives $Aq \subseteq A\mu$.
2. The source's sufficient criterion is applied to the common ideal using the norm-one generator q, not to the norm-two generator μ.
3. The density clause cannot be rescued by choosing another density: Radon–Nikodym densities with respect to a fixed Haar measure are unique, and their L^1 norm equals total variation.
4. All coefficients and convolution identities in the S_3 example were checked with exact rational arithmetic; no floating-point or convention- sensitive inference is used.

Bounded novelty check

On 11 August 2026, the arXiv id, exact conjecture, idempotent measures, principal left ideals, and metric projectivity were checked in the four run indexes and by bounded exact-phrase web/arXiv searches. The conjecture also appears unchanged in the recent published version located by the search. No later resolution or this S_3 counterexample was found. This is evidence for novelty, not a definitive guarantee.

References

[1] N. Nemesh, *Metric and topological homology. Projectivity, injectivity and flatness*, arXiv:2203.09765, 2022.